

Laboratoriya 10

Mavzu: CSS transition va animatsiya elementlarini yaratish

1. Kirish qismi

Zamonaviy veb-saytlar foydalanuvchiga qulay, dinamik va interaktiv bo'lishi uchun turli vizual effektlardan keng foydalanadi. Bunday effektlar ichida CSS transition (o'tishlar) va CSS animation (animatsiyalar) eng ko'p qo'llaniladigan vositalardan hisoblanadi. Ular yordamida elementlarning harakatlanishi, hajmi o'zgarishi, ranglar almashishi yoki boshqa ko'plab o'zgarishlar silliq va bosqichma-bosqich amalga oshiriladi. JavaScriptdan foydalanmasdan ham murakkab animatsion effektlar yaratish imkoniyati loyihalash jarayonini soddalashtiradi hamda sahifaning vizual ko'rinishini yaxshilaydi.

Ushbu laboratoriya ishida talabalarning CSS transition va animation asoslarini o'zlashtirishlari, oddiydan tortib murakkabgacha bo'lgan animatsiyalarni mustaqil yaratish ko'nikmalarini shakllantirish maqsad qilingan.

Ishning maqsadi

CSS transition tushunchasi va uning asosiy xususiyatlari bilan tanishish.

Transition yordamida element xossalarini silliq o'zgartirishni amaliy o'rganish.

CSS animation (keyframes animatsiya) mexanizmini o'zlashtirish.

Turli animatsion effektlar yaratish, ularni optimallashtirish va sahifaga qo'llash.

Interaktiv veb elementlar loyihalashda animatsiyalarni samarali qo'llash bo'yicha ko'nikma hosil qilish.

3. Nazariy qism

3.1. CSS Transition tushunchasi

CSS transition — elementning biror bir CSS xossasi (masalan, rang, o'lcham, pozitsiya, shaffoflik) o'zgarishi mobaynida bu o'zgarishni belgilangan vaqt davomida silliq amalga oshirish jarayonidir.

Transitionning asosiy maqsadi — statik o'zgarishlarni ko'rinarli, yumshoq va chiroyli holga keltirish.

Transitionni qo'llash uchun quyidagi asosiy xossalar ishlatiladi:

1) transition-property: Qaysi xossa o'zgarayotganini bildiradi.

Misol: transition-property: width;

2) transition-duration: O'tish davomiyligini belgilaydi (sekundlarda).

Misol: transition-duration: 0.5s;

3) transition-timing-function: O'tish tezligi qanday o'zgarishini ko'rsatadi.

Asosiy qiymatlar:(ease , linear , ease-in , ease-out , ease-in-out , cubic-bezier(x, y, z, t))

4) transition-delay

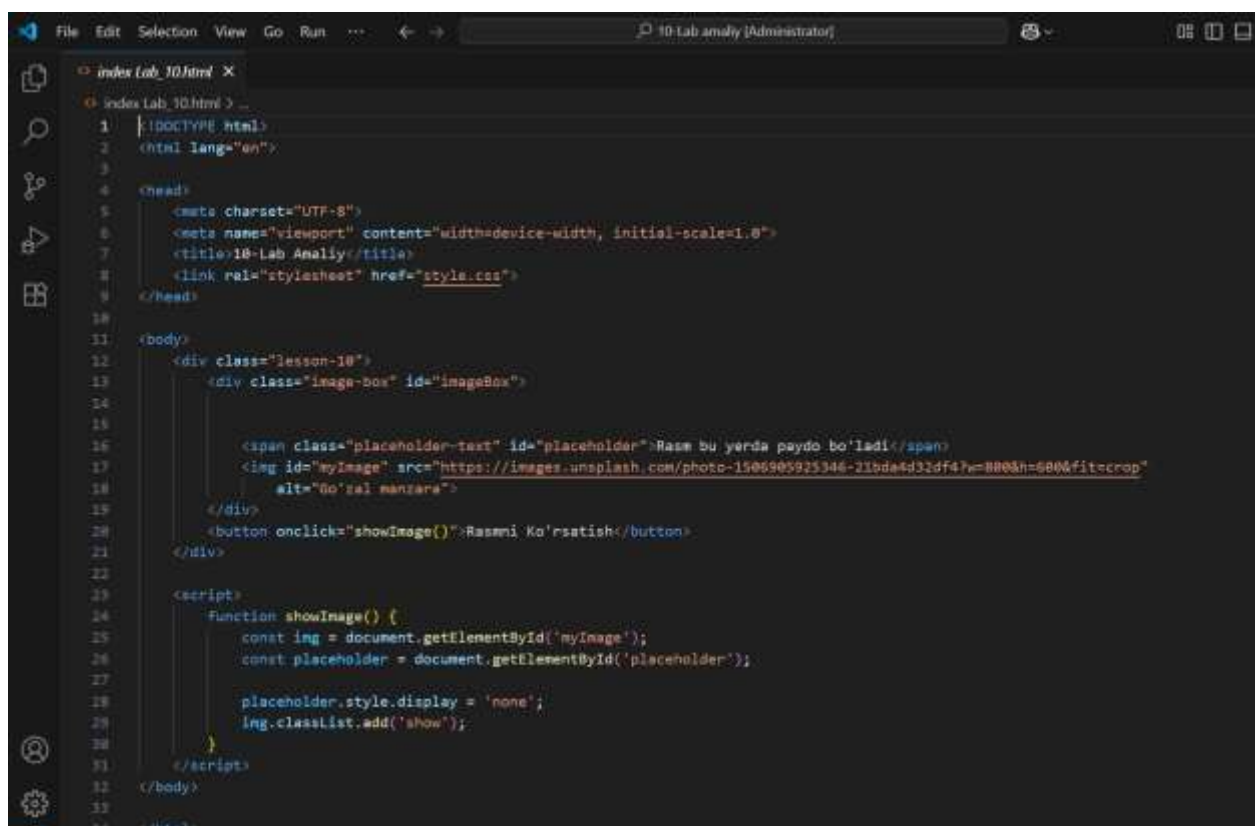
O'tishni boshlanishidan oldin qancha kutish kerakligini bildiradi.

Transitionning umumiy yozilishi:

transition: property duration timing-function delay;

Amali qism:

html kod:

A screenshot of a code editor window titled 'index Lab_10.html'. The editor shows HTML code with a head section containing meta tags for charset, viewport, and title, and a link to a stylesheet. The body section contains a div with class 'lesson-10' and id 'imageBox', which includes a placeholder text, an image with a specific URL, and a button with an onclick event 'showImage()'. Below the HTML, there is a JavaScript function 'showImage()' that uses 'document.getElementById' to find the image and placeholder elements, hides the placeholder, and adds a 'show' class to the image.

```
1 |<!DOCTYPE html>
2 |<html lang="en">
3 |
4 |<head>
5 |   <meta charset="UTF-8">
6 |   <meta name="viewport" content="width=device-width, initial-scale=1.0">
7 |   <title>10-Lab Analizy</title>
8 |   <link rel="stylesheet" href="style.css">
9 |</head>
10 |
11 |<body>
12 |   <div class="lesson-10">
13 |     <div class="image-box" id="imageBox">
14 |
15 |
16 |       <span class="placeholder-text" id="placeholder">Rasm bu yerda paydo bo'ladi</span>
17 |       
19 |     </div>
20 |     <button onclick="showImage()">Rasmni Ko'rsatish</button>
21 |   </div>
22 |
23 |   <script>
24 |     function showImage() {
25 |       const img = document.getElementById('myImage');
26 |       const placeholder = document.getElementById('placeholder');
27 |
28 |       placeholder.style.display = 'none';
29 |       img.classList.add('show');
30 |     }
31 |   </script>
32 |</body>
33 |
34 |</html>
```

CSS kod:

```
body {
```

```

font-family: Arial, sans-serif;
display: flex;
flex-direction: column;
align-items: center;
justify-content: center;
min-height: 100vh;
margin: 0;
background: linear-gradient(135deg, #0f8b19 0%, #dfe76c 100%);
}

.lesson-10 {
  text-align: center;
  background: white;
  padding: 40px;
  border-radius: 20px;
  box-shadow: 0 10px 40px rgba(0, 0, 0, 0.2);
}

.image-box {
  width: 500px;
  height: 500px;
  border: 3px dashed #c91d1d;
  border-radius: 10px;
  margin-bottom: 20px;
  display: flex;
  align-items: center;
  justify-content: center;
  background: #f9f9f9;
  overflow: hidden;
}

.image-box img {
  max-width: 500px;
  max-height: 500px;
  display: none;
  border-radius: 8px;
}

.image-box img.show {
  display: block;
  animation: fadeIn 0.5s ease-in;
}

@keyframes fadeIn {
  from {
    opacity: 0;
    transform: scale(0.8);
  }

  to {
    opacity: 1;

```

```
        transform: scale(1);
    }
}

button {
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    color: white;
    border: none;
    padding: 15px 40px;
    font-size: 18px;
    border-radius: 50px;
    cursor: pointer;
    transition: all 0.3s ease;
    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);
    display: flex;
    align-items: center;
    justify-content: center;
    animation: horizontalMove 2s infinite ease-in-out;
}

button:hover {
    transform: translateY(-2px);
    box-shadow: 0 6px 20px rgba(0, 0, 0, 0.3);
}

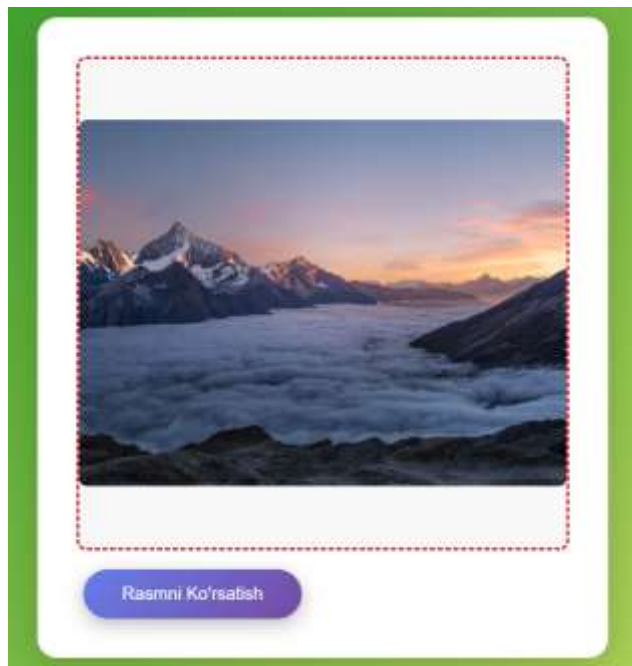
button:active {
    transform: translateY(0);
}

@keyframes horizontalMove {
    0%,
    100% {
        transform: translateX(0);
    }

    50% {
        transform: translateX(10px);
    }
}

.placeholder-text {
    color: #e74100;
    font-size: 18px;
}
```

Natija:



Bu yerda buttonga animatsiya qoshilgan.

Xulosa:

Bu loyiha quyidagi asosiy funksiyalarni amalga oshiradi:

- Sahifada **.image-box** nomli blok mavjud bo'lib, u rasm uchun joy ajratadi.
- Ichida **Rasm bu yerda paydo bo'ladi** nomli joy tutuvchi matn (**placeholder-text**) va asl rasm (****) bor.
- **showImage()** funksiyasini chaqiradigan **Rasmni Ko'rsatish** tugmasi joylashtirilgan.
- Rasm dastlab CSS orqali **yashirilgan** holatda turadi.
- **body** uchun jozibali, yashil-sariq gradient fon o'rnatilgan.
- **.image-box** o'ziga xos qizil, punktir (dashed) chegara va sobit o'lchamga (500x500px) ega.
- Tugma chiroyli, binafsha gradient fon, yumaloq qirralar va ikki xil animatsiyaga ega:
 - **horizontalMove**: Tugmani gorizontaal ravishda sekin harakatga keltirib turadi.

- **hover**: Sichqoncha ustiga borganda biroz yuqoriga siljiydi (kinetik effekt).
- **.image-box img** odatda **display: none** orqali yashirilgan.
- **.image-box img.show** sinfi qo'shilganda esa **display: block** bo'ladi va **fadeIn** animatsiyasi yordamida silliq paydo bo'ladi (kattalashib va xiralikdan chiqib).
- **showImage()** funksiyasi quyidagilarni amalga oshiradi:
 1. Joy tutuvchi matnni (placeholder) **yashiradi** (display = 'none').
 2. Rasm elementiga **show** CSS sinfini qo'shadi.
- Natijada rasm paydo bo'ladi va **fadeIn** animatsiyasi ishga tushadi.